

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 4.2: Probability I — Teacher Reference (Pre-filled)

SUMMARY TABLE: MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 4.2: Probability I
Driving Question	DRIVING QUESTION: How is probability applied in real life situations?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Experimental Probability I: Coin Tosses and the 70% Forecast	We tossed a coin or bottle top 10 times in groups (Heads = rain, Tails = no rain) and recorded the results in a tally table, e.g. 6 heads and 4 tails. Different groups got different results. We also saw that Kericho had 14 rainy days out of 20 in past March records.	Probability is a number from 0 (impossible) to 1 (certain) that tells us how likely an event is. Relative frequency (experimental probability) = number of times the event occurred / total number of trials, e.g. $6/10 = 0.6$ and $14/20 = 0.7$. A single trial cannot be predicted for certain.	A forecast of 70% chance of rain does not guarantee rain. It is a likelihood estimated from past records, meaning that about 7 out of every 10 similar days have rain. Wanjiru, Captain Otieno and Ms Achieng can use such numbers to make decisions, but we still need to find out how the numbers are worked out and combined.
Lesson 2 Experimental Probability II: More Trials, Pooled Data and the Kericho Records	We pooled results from 8 groups (80 tosses, 42 heads and 38 tails) and watched the running relative frequency of heads change: 10 tosses, 0.6; 20 tosses, 0.55; 40 tosses, 0.55; 80 tosses, 0.525. The Kericho March record showed rain on 14 days out of 20, so $14/20 = 0.7$.	Relative frequency = number of times the event occurred / total number of trials. As the number of trials increases, relative frequency becomes steadier and gives a better estimate of probability. Pooled data and long records (like Kericho) give more reliable estimates than small samples.	A forecast of 70% chance of rain is based on a long record of similar days (about 7 in every 10 had rain), so it is a trustworthy estimate. It does not guarantee rain on one day, but it helps Wanjiru, Captain Otieno and the county office make sensible decisions over many days.
Lesson 3 Probability Space I: Listing Sample Spaces and Defining Events	What we observed: In our coin, two-coin and die activities, we saw these outcomes: one coin {H, T}; two coins {HH, HT, TH, TT} (HT and TH are different); die {1, 2, 3, 4, 5, 6}. A weather day on the KMD card has three states: rain, storm, dry.	What we learned: A sample space S is the complete list of all possible outcomes of a random experiment. An event is a subset of S, for example 'even number' = {2, 4, 6}, and 'rain, as the farmer sees it' = {rain, storm}. An event can have one outcome, several outcomes or all outcomes of S.	How this helps explain the phenomenon: Before KMD can say '70% chance of rain', it must list all possible day states $S = \{\text{rain, storm, dry}\}$ and decide which outcomes make up the event. Wanjiru, Captain Otieno and Ms Achieng each care about a different event within the same sample space. We still need to find how likely each outcome is (Lesson 4).

Lesson 4 Probability Space II: Theoretical Probability and Why It Differs from What We Observe	We observed that our class pooled coin result ($42/80 = 0.525$) and our die results were close to, but not exactly equal to, the theoretical values of 0.5 for a fair coin and $1/2$ for $P(\text{even})$ on a die. We also saw that the weather states cannot be counted as equally likely, since $P(\text{rain only}) = 0.6$, $P(\text{storm}) = 0.1$ and $P(\text{dry}) = 0.3$ add up to 1.0.	We learned that theoretical probability = favourable outcomes / total equally likely outcomes, and that experimental probability = relative frequency from trials or records. The two usually differ a little because of chance variation, and the difference tends to shrink as the number of trials grows.	A forecast of 70% rain does not guarantee rain because probabilities describe the long run, not one day. For weather, the numbers come from records such as Kericho's 14 rainy days out of 20 and from models, because rain, storm and dry are not equally likely, so Wanjiru, Captain Otieno and Ms Achieng can trust the numbers even though a single day may turn out differently.
Lesson 5 Mutually Exclusive and Independent Events: Can Rain and Sunshine Happen Together?	What did you observe? For rain and dry on the same day, did any outcome appear in both events? For rain on Monday and rain on Tuesday (and for the first and second coin toss), did the chance of the second change after the first? For die even and coin Heads, and for die even and die shows 2, what did you find?	What did you learn? Mutually exclusive events cannot happen together, so $P(A \text{ and } B) = 0$. Independent events do not change each other's chance. Some pairs are neither. These are two different questions.	How does this help explain the 70% forecast puzzle? Explain why Wanjiru, Captain Otieno and Ms Achieng must first decide whether events are mutually exclusive, independent or neither before combining probabilities such as $P(\text{rain only}) = 0.6$, $P(\text{storm}) = 0.1$ and $P(\text{dry}) = 0.3$, or rain on two days.
Lesson 6 Laws of Probability I: The Addition Law and the Complement Rule	We observed that on the grid of 100 similar March days in Kericho, 60 days were rain only, 10 were storm and 30 were dry; the days that were 'rain only or storm' numbered 70, the days that were 'storm or dry' numbered 40, and the days with no rain numbered 30. No day was counted in two states at once.	We learned that for mutually exclusive events, $P(A \text{ or } B) = P(A) + P(B)$, and that $P(\text{not } A) = 1 - P(A)$. So $P(\text{rain only or storm}) = 0.6 + 0.1 = 0.7$, $P(\text{storm or dry}) = 0.1 + 0.3 = 0.4$, and $P(\text{no rain}) = 1 - 0.7 = 0.3$.	This helps explain the forecast because the 70% chance of rain is made up of rain only (0.6) and storm (0.1), and the 30% chance of no rain is what remains. Farmer Wanjiru, Captain Otieno and the county office can use these numbers to compare the chances with their cost of being wrong, even though the forecast does not guarantee what will happen on one day.
Lesson 7 Laws of Probability II: The Multiplication Law and Two Rainy Days	We tossed a coin and rolled a die together and recorded how often we got a head and an even number. We also wrote the four outcomes for two days of weather (RR, RN, NR, NN) with $P(R) = 0.7$ and $P(N) = 0.3$ each day.	For independent events, $P(A \text{ and } B) = P(A) \times P(B)$. The chance of rain on both days is $0.7 \times 0.7 = 0.49$ and the chance of no rain on both days is $0.3 \times 0.3 = 0.09$. The chance of both events is smaller than the chance of either one alone.	Wanjiru Chebet can use the multiplication law to turn the KMD one-day forecast of 0.7 into a two-day chance of 0.49. This is not a guarantee, but it tells her that about 49 out of every 100 such pairs of days have rain on both days, which helps her decide whether to plant, wait or irrigate.
Lesson 8 Probability Tree Diagrams: Mapping Two Days of Weather	We drew a two-day tree for rain (R, 0.7) and no rain (N, 0.3) and found the four outcomes RR, RN, NR, NN. Multiplying along the branches gave 0.49, 0.21, 0.21 and 0.09, and these add to 1.00.	A tree diagram shows all outcomes of combined events. We multiply along branches (multiplication law) and add the outcomes that make an event (addition law). Exactly one rainy day = $0.21 + 0.21 = 0.42$, and at least one rainy day = $0.49 + 0.21 + 0.21 = 0.91$, which equals $1 - 0.09$.	Even though the forecast for one day is 0.7, the chance of at least one rainy day in two is higher, 0.91, because there are more ways to get rain. The tree lets Wanjiru, Captain Otieno and the county office combine probabilities, but the results are long-run chances, so a single pair of days can still be dry, with probability 0.09.
Lesson 9 Applying probability to	We observed that the forecast of 70% rain in Kericho matches the past record of 14	We learned that probability is a number from 0 to 1 that measures how likely an	A forecast of 70% chance of rain does not guarantee rain. It means that in a long run

real decisions: advising the farmer, the pilot and the county office	rainy days out of 20 ($14/20 = 0.7$), that the day states have probabilities rain only 0.6, storm 0.1 and dry 0.3 (sum 1.0), and that the two-day tree gives RR 0.49, RN 0.21, NR 0.21, NN 0.09 (total 1.00), with at least one rainy day 0.91 and exactly one 0.42. Comparing these with the class thresholds led to different decisions for Wanjiru, Captain Otieno and Ms Achieng.	event is. It can be estimated from relative frequency or found by theoretical probability. We learned to list sample spaces, to use the addition law for mutually exclusive events, the multiplication law for independent events, the complement rule and tree diagrams. We also learned to compare probabilities with a decision threshold and with the cost of being wrong.	of similar days about 7 out of every 10 have rain, worked out from past records and models. Because these numbers come from many past trials, Wanjiru can prepare to plant and keep irrigation as backup, Captain Otieno can plan for a possible 10% storm and check again before take-off, and the county office can prepare drainage and warn residents. A decision based on probability may be unlucky on one day, but it is the best choice over many days. This is how probability is applied in real life situations.
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ATTRIBUTION AND LICENSING

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